

For 4% Initial Spending Rate, 50/50 Asset Allocation, Rolling 30-Year Retirements

Using SBBI Data, 1926–2016, S&P 500 and Intermediate-Term Government Bonds

Retirement Date Wealth Level = \$100,000

Spending Strategy			Real	Real	Real	Real	Change	Year 30	
			Spending	Spending	Spending	Remaining	in Real	Spending	
	Initial	Percentile	in 10	in 20	in 30	Wealth	Spending	Relative	Downside
	Spending Rate	of Distribution	Years	Years	Years	After 30 Years	at Year 30	to Baseline	Spending Volatility
Constant Inflation-Adjusted Spending (BASELINE)	4.00%	90th	\$4,000	\$4,000	\$4,000	\$322,310	0%	0%	0%
		50th	\$4,000	\$4,000	\$4,000	\$127,010	0%	0%	0%
		10th	\$4,000	\$4,000	\$4,000	\$27,830	0%	0%	0%
Fixed Percentage Withdrawals	4.00%	90th	\$6,010	\$7,730	\$7,760	\$202,690	94%	94%	-3%
		50th	\$4,170	\$4,440	\$4,800	\$120,280	20%	20%	-4%
		10th	\$2,570	\$2,420	\$3,250	\$87,480	-19%	-19%	-5%
Endowment Formula—Hybrid	4.00%	90th	\$5,100	\$6,470	\$6,870	\$256,830	72%	72%	-2%
		50th	\$4,090	\$4,180	\$4,340	\$124,270	9%	9%	-2%
		10th	\$3,230	\$3,160	\$3,380	\$73,200	-16%	-16%	-3%
Endowment Formula—3 Year Moving Average	4.00%	90th	\$5,100	\$6,560	\$7,110	\$186,160	78%	78%	-2%
		50th	\$3,860	\$4,110	\$4,460	\$120,040	12%	12%	-2%
		10th	\$2,580	\$2,230	\$2,900	\$84,940	-28%	-28%	-3%
Bengen's Dollar Floor-and-Ceiling Withdrawals	4.00%	90th	\$4,800	\$4,800	\$4,800	\$288,960	20%	20%	0%
		50th	\$4,170	\$4,400	\$4,760	\$122,810	19%	19%	-1%
		10th	\$3,400	\$3,400	\$3,400	\$68,760	-15%	-15%	-3%
Vanguard's Percentage Floor-and-Ceiling Withdrawals	4.00%	90th	\$5,130	\$5,780	\$6,420	\$318,970	61%	61%	0%
		50th	\$3,820	\$3,570	\$4,230	\$132,200	6%	6%	-2%
		10th	\$2,560	\$2,070	\$2,240	\$94,920	-44%	-44%	-3%
Kitces Ratcheting Rule	4.00%	90th	\$4,840	\$6,440	\$9,430	\$231,710	136%	136%	0%
		50th	\$4,000	\$5,320	\$7,090	\$81,430	77%	77%	0%
		10th	\$4,000	\$4,000	\$4,000	\$25,900	0%	0%	0%